Dividing Complex Numbers

Dividing complex numbers by multiplying by the conjugate

Directions

- A quotient is not in standard form while an i remains in the denominator. Multiply the top and the bottom by the *conjugate* of the denominator, then simplify.
- The conjugate of $c + di$ is $c - di$; the conjugate of ci is $-ci$.
- Write every final answer as $a + bi$ with a and b real. Show the multiplied-out numerator and denominator before you divide.
- In the applied problems, the quantities and their units are given in the stem; give the answer in the unit printed on the answer line.

1. Simplify: $\dfrac{8+6i}{2i}$

Answer: ________________

2. Simplify: $\dfrac{50}{3+4i}$

Answer: ________________

3. Simplify: $\dfrac{9-12i}{3i}$

Answer: ________________

4. Simplify: $\dfrac{16+11i}{2+3i}$

Answer: ________________

5. A steady alternating-current source supplies a voltage of 20 volts across a component whose impedance is $3+i$ ohms. Ohm's law for this circuit says the current, in amps, equals the voltage divided by the impedance. Find the current.

Answer: ________________ amps

6. Simplify: $\dfrac{-8+6i}{1-2i}$

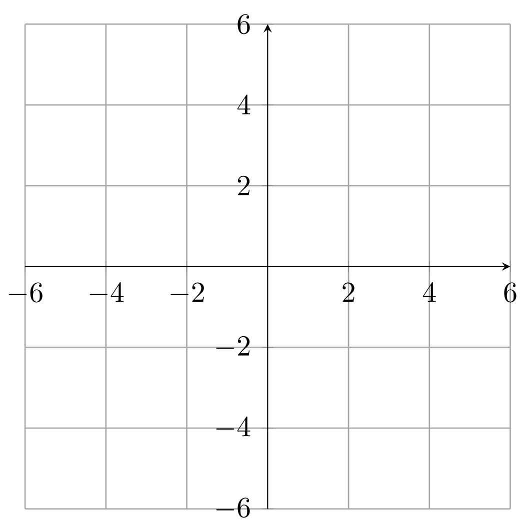
Answer: ________________

7. A drawing program transforms a point of the complex plane by multiplying it by $3+4i$. One transformed point landed at $-7+24i$.

(a) Which point did the program start from? Divide to undo the multiplication.

Answer: ____________________

(b) Plot your answer to part (a) on the plane below and label the two axes as the real axis and the imaginary axis.



8. Across one component of an alternating-current circuit the voltage is $26 + 13i$ volts and the current through it is $2 + 3i$ amps. The impedance, in ohms, equals the voltage divided by the current. Find the impedance.

Answer: ________________ ohms

9. Simplify: $\dfrac{29i}{5 + 2i}$

Answer: ________________

10. Consider the sum $\dfrac{3+i}{1-i} + \dfrac{3-i}{1+i}$.

(a) Simplify each quotient separately, then add. Write the result in standard form.

Answer: ________________

(b) The result is a real number. In one or two sentences, justify why that had to happen, using the relationship between the two quotients.

Answer: ________________